

Photosynthesis and Energy

Photosynthesis is the biochemical process by which green plants, algae and cyanobacteria convert light energy into chemical energy stored in glucose, using water and carbon dioxide while releasing oxygen.

The light-dependent reactions occur in the thylakoid membranes, where chlorophyll absorbs photons and splits water molecules. The Calvin cycle, discovered by Melvin Calvin in 1950, fixes carbon dioxide into three-carbon sugars using the ATP and NADPH produced earlier.

Photosynthesis produces approximately 100 billion tons of biomass every year, sustaining nearly all life on Earth.